

EduNexus

Advanced Learning Management System (LMS)



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Outline

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- EduNexus is a MERN-based LMS platform.
- Targets both academic and career-focused users.
- Students and instructors are primary stakeholders.
- Offers scalable learning and assessment tools.
- Aims to improve engagement and career readiness.

Project Objectives

- Enable course creation and structured learning flow
- Automate quizzes with AI to promote smart assessment
- Offer tools for resume and cover letter generation
- Ensure secure transactions using Razorpay integration
- Promote a role-based, modular, and scalable system

System Requirements & Technology Stack

- **Frontend:** React.js, Tailwind CSS, Redux Toolkit
- **Backend:** Node.js and Express.js
- **Database:** MongoDB with Mongoose
- **Authentication:** JWT, Bcrypt, OTP via Nodemailer
- **AI Features:** Gemini API (quizzes, cover letters)

Functional Requirements: Key Capabilities

User Management:

- Email OTP registration, JWT-secured login
- Role-based access (Student, Instructor, Admin)
- Profile updates, avatar, bio

Course Management:

- Create structured courses, content uploads
- Student filtering, preview, enrollment

Transactions:

- Razorpay payment, backend verification
- Automated enrollment confirmation

Functional Requirements: Unique Tools & Features (1)

Quiz Generator:

- Students input topic, number of questions and level of questions
- Gemini API generates MCQs with explanations
- Interactive interface, results visualization
- Promotes self-assessment

Resume Builder:

- Form-based input (skills, education, etc.)
- Converts to PDF via html2pdf
- Includes preview option

Functional Requirements: Unique Tools & Features (2)

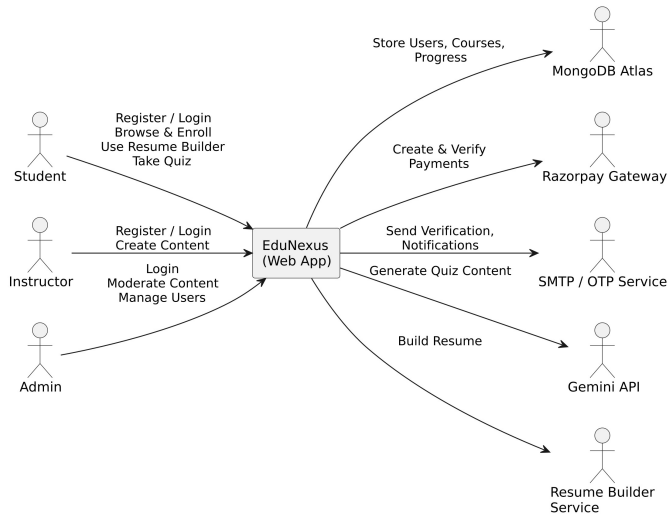
Cover Letter Generator:

- Gemini API-based (job role, company, desc.)
- Outputs editable, downloadable letters
- Supports career growth

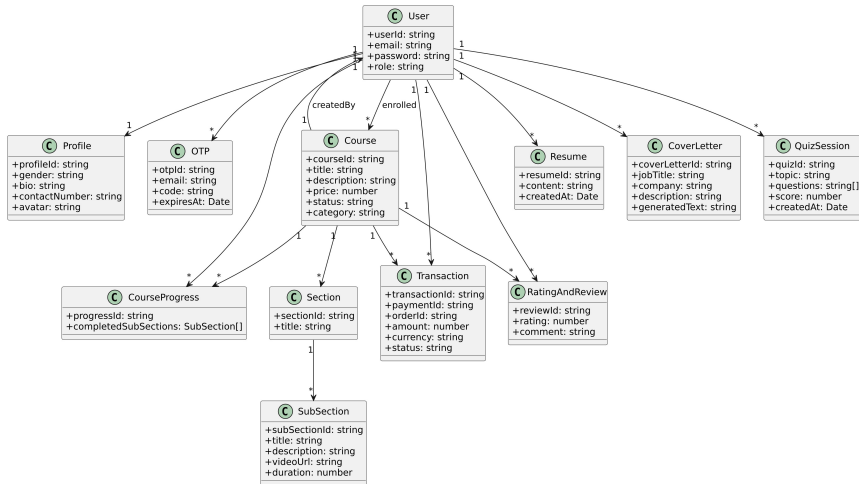
Rating & Review System:

- Course reviews with star ratings and feedback
- Improves content quality and trust

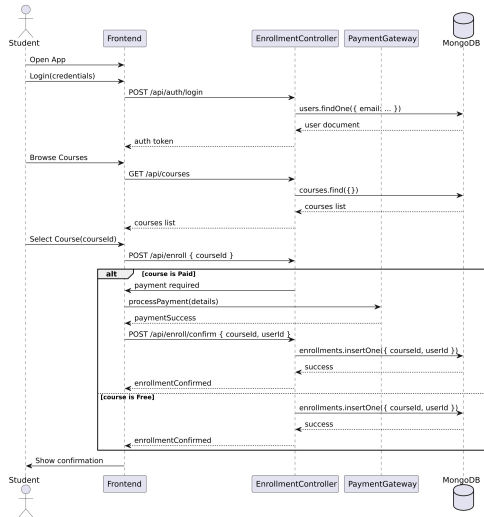
High-Level System Design



Class Diagram



Sequence Diagram



- **Students:** Enroll, Learn, Review
- **Instructors:** Create courses and Manage course
- **Admins:** Manage users, content, reviews

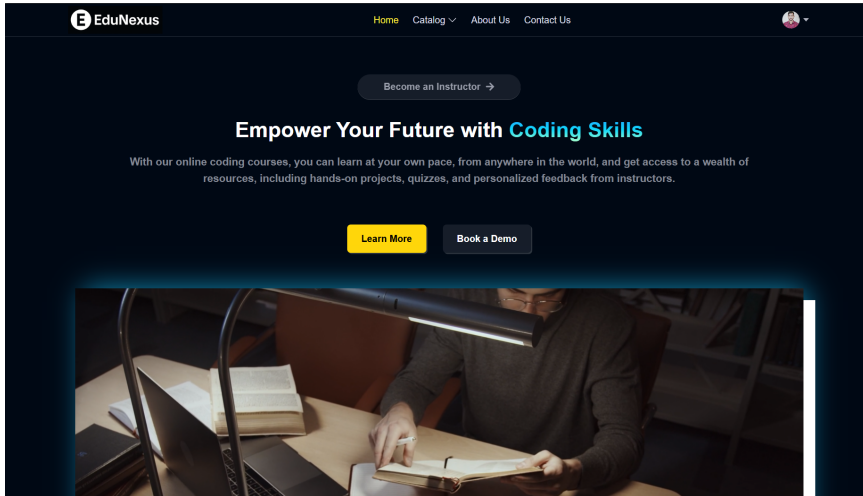
Implementation

- JWT, Bcrypt for secure authentication
- Nodemailer for OTP-based registration
- MongoDB for course and user data, Cloudinary for user media content
- Gemini API for quizzes and cover letters
- html2pdf to generate resume PDFs
- Razorpay SDK integration for secure payments
- Admin endpoints for content moderation
- Responsive UI with React.js + Tailwind CSS

- Manual and iterative testing cycles
- Unit, integration, and UAT performed
- Tools: Postman, Browser DevTools
- Covered API, UI, integration, and security

Test Results

- Registration, login, and verification successful
- Payments correctly processed and verified
- Quiz generation via Gemini API functional
- Cover letter creation using Gemini API working as expected
- Resume generation and PDF export executed correctly



The screenshot shows the 'My Profile' page in the EduNexus application. The interface has a dark theme. At the top, there is a navigation bar with the EduNexus logo on the left and links for Home, Catalog, About Us, and Contact Us on the right. A user profile icon with the initials 'AR' is in the top right corner. On the left side, there is a sidebar menu with options: My Profile (highlighted), Dashboard, My Courses, Add Course, Settings, and Logout. The main content area is titled 'My Profile' and contains two sections. The first section displays the user's profile information: a circular profile picture with 'AR', the name 'Abhishek Rao', and the email 'abhirao274408@gmail.com'. To the right of this information is a yellow 'Edit' button with a pencil icon. The second section is titled 'About' and contains the text 'BSc :- BHU and MCA :- MNNIT'. To the right of this text is another yellow 'Edit' button with a pencil icon.

EduNexus Home Catalog About Us Contact Us AR

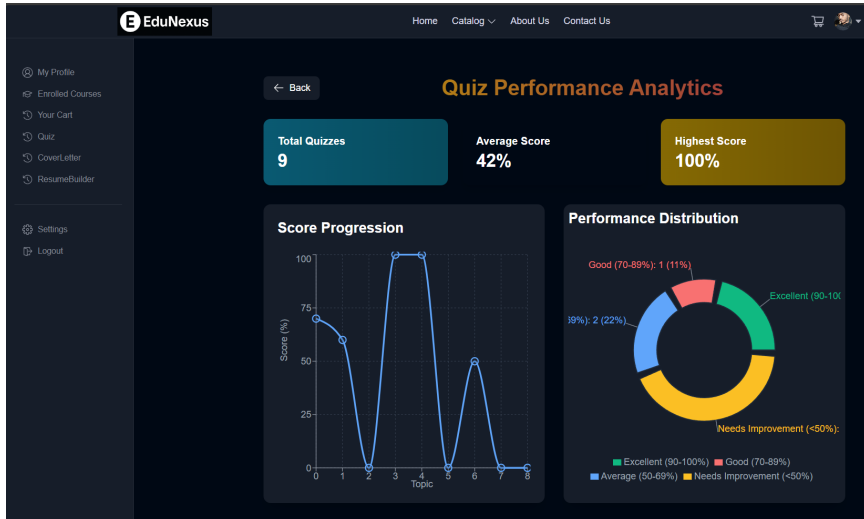
My Profile

AR **Abhishek Rao**
abhirao274408@gmail.com **Edit**

About **Edit**

BSc :- BHU and MCA :- MNNIT

Quiz: Quiz result



- Mobile app (PWA/Hybrid)
- Live sessions and bootcamps
- Auto-evaluation and certifications
- Forums for community interaction
- Gamified learning (badges, points)
- Admin moderation tools and analytics
- AI-based personalized learning

Conclusion

- Built MERN-based LMS platform with career tools
- Solves problems like fragmented tools and lack of interactivity
- Includes secure course access and payments
- Uses Gemini API for AI-powered features
- Implements core LMS modules with role-based access
- Fully tested for usability and reliability

- MERN Stack Documentation – React.js Docs
- Node.js Documentation – Node.js Docs
- MongoDB Documentation – MongoDB Docs
- Razorpay Developer Docs – Razorpay Docs
- html2pdf.js – html2pdf.js GitHub

Questions?

Thank you for your attention!